

IPM 2026/27 · Practical project

Centro de Operações do Lima (COL)

Project statement. Human-Machine Interaction, DCC/FCUP. Álvaro Figueira and Diogo Abrantes.

This document describes the single project of the whole class: **a graphical interface for the operations room of a stretch of river with two dams**. All groups work on the same problem. What sets groups apart is the quality of their decisions, the way they justify them, and each group's "gold" (section 6). You do not need to understand everything at first reading. Technical terms are explained as they appear, and there is a glossary at the end.

Key dates

When	What
29 Sep (S2)	Project start: this statement is read and discussed in the practical class and in the lecture.
Week of 3 Nov (S6)	Evaluation phase 1: cognitive walkthrough (see section 8), done in class, between groups.
Sunday 8 Nov, 23:30	Report 1 (R1) due, in PDF.
Week of 17 Nov (S8)	Evaluation phase 2: heuristic evaluation (see section 8), done in class, between groups.
Week of 24 Nov (S9)	Evaluation phase 3: usability tests (see section 8), in class and then with 5 people from outside.
Friday 4 Dec	Status report from each group, by email, to the lecturers.
Sunday 13 Dec, 23:30	Report 2 (PR2) due, with the prototype in Figma (the design tool you will use from S7) and the records of the three phases.
Tuesday 15 Dec	Final presentations: P1 and P2 from 9:00 to 11:00, P3 from 11:00 to 13:00.

1. Description

The river Lima rises in Galicia and enters Portugal at Ponte da Barca. From there to the sea, at Viana do Castelo, it runs for about 60 km. On that stretch there are two **dams**. The water held back behind each dam forms an artificial lake, the **reservoir** (in Portuguese, albufeira). A dam has openings that can be opened and closed, so water only passes when someone decides to open the **gates**.

The first dam, **Alto Lindoso**, is the country's largest producer of electricity from water. Water from the reservoir drops through the inside of the mountain and spins turbines, which generate electricity. In this document, sending water through the turbines is called **turbining** (turbinar). Alto Lindoso does not turbine continuously. It produces when the electricity grid needs more energy, for example in the early evening, and stops when it does not. The result is that, a few kilometres further down the river, the Lima receives large surges of water for a few hours and almost nothing for the rest of the day.

The second dam, **Touvedo**, is 12 km further down and exists precisely to solve that problem. It is much smaller, and its job is to receive the surges from Lindoso and return them to the river evenly, without peaks. But Touvedo's reservoir is small. When it rains hard, the smaller rivers that flow into it and the rain itself fill it quickly, and Touvedo can no longer hold what it receives. It then has to **release water** (largar água): open the gates (the large steel doors at the top of the dam) and let part of what arrives pass into the river.

The amount of water passing along a river, its **flow** (caudal), is measured in **cubic metres per second (m³/s)**. One cubic metre is 1000 litres. When we say that Touvedo is releasing 500 m³/s, that means 500

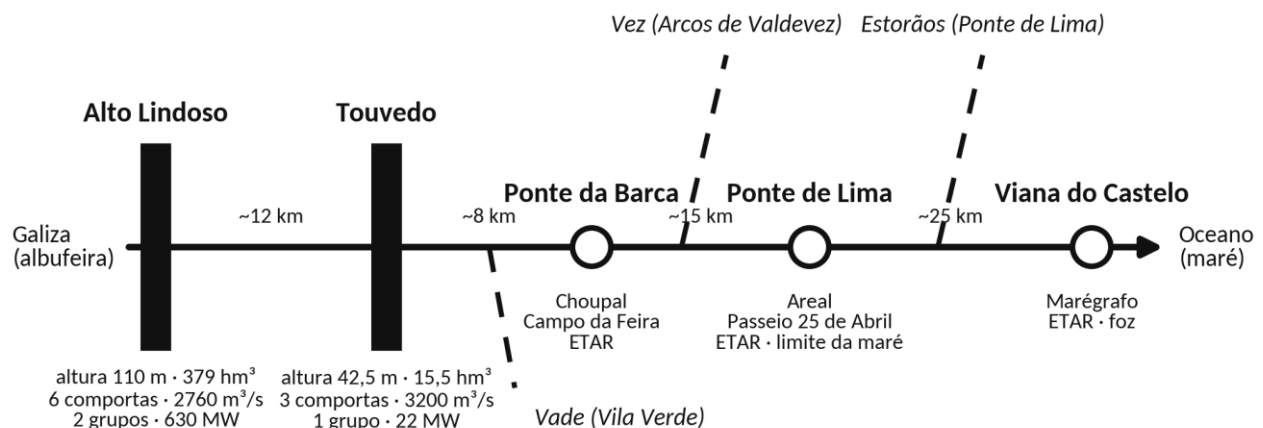
thousand litres per second, or an Olympic swimming pool every five seconds. In dry weather the Lima carries well under 100 m³/s at Ponte de Lima.

There are also two factors that nobody controls. The first is the **tributaries** (afluentes), the smaller rivers that flow into the Lima. The ones that matter are the Vez, the Vade and the Estorãos. They have no dams, so when it rains they bring water into the Lima and there is no way to hold it back. The second is the **tide**. Near the sea, the level of the river rises and falls twice a day with the tide, and this effect reaches as far as Ponte de Lima. At **high tide** (preia-mar, the highest point of the tide) the river drains more slowly, and the water coming from upriver builds up.

What happened on 2 February 2026

That day, after several days of heavy rain, Touvedo was releasing about 500 m³/s at 10 in the morning. The Vez and the Vade were running full. High tide at Ponte de Lima was at 15:00. The river covered the sandy riverbank of Ponte de Lima (the areal), reached the wall of the Passeio 25 de Abril (the riverside promenade), entered the Clube Náutico and, at Ponte da Barca, overflowed the banks at the Choupal (a riverside park). There were no victims, but there was heavy damage.

Today, the water releases are decided by EDP (the company that owns the dams). Warning people is the job of APA (the public agency that manages rivers) and of the Civil Protection. Responding on the ground is the job of the fire brigades of the two municipalities. These bodies sit in different places and talk by phone. **Your job is to design the graphical interface of a single operations room for this stretch of river, which we will call the Centro de Operações do Lima (COL).** The COL is fictional. It brings the three functions into one place: deciding what the dams do, warning those at risk and coordinating the teams on the ground. The data you will use are invented, but the dams, the towns, the tributaries and the tide are real.



Esquema do rio Lima entre a fronteira e o mar, não à escala. Distâncias aproximadas em linha reta. Dados das barragens: CNPGB/APA.

Legenda

- Barragem: parede que retém a água (a albufeira) e que a pode largar de forma controlada
- Vila ou cidade junto ao rio (por baixo do nome, os locais que o COL vigia)
- Afluente sem controlo: rio que entra no Lima e cuja água ninguém consegue reter
- Sentido da corrente: de montante (a Galiza) para jusante (o mar)

Figure 1. The stretch of the Lima that the COL controls: two dams, three uncontrolled tributaries, three towns and the tide.

2. Who will use the interface

The COL has **three operators**, one per 8-hour shift, and a **coordinator** who is not in the room but can be reached at any hour. From October to April, the rainy season, the room works 24 hours a day. Outside that period it works only during the day. The interface is designed for the operator. The coordinator should appear in the interface only as the person who authorises certain decisions.

The operator can do the following on his or her own decision:

- **Acknowledge an alert**, that is, mark that the alert has been seen and that he or she is now responsible for it.
- **Change the turbine flow**, that is, increase or reduce the water going through the turbines. This changes the electricity produced and also the flow released into the river.
- **Open or close the bottom outlet** (descarga de fundo), a pipe with a valve near the base of the dam, through which water can be released without going through the turbines or the gates at the top.
- **Move the gates** at Touvedo, as long as the total water released by the dam stays under 300 m³/s.
- **Request water analyses** from a station and **send field teams** to a place.

For what follows, the operator needs the **coordinator's authorisation**, requested and received through the interface:

- Releases above 300 m³/s.
- Warnings to the population.
- Closing bridges, parks or the Passeio 25 de Abril.

Every shift ends with a recorded **shift handover** (passagem de turno).

3. The system that exists on the ground

Everything in this section is given. You may not add dams, sensors or teams, and you may not remove any. The figures for the dams are real.

The two dams

Alto Lindoso dam (Ponte da Barca, 1992). It is an **arch dam**: the wall is curved, like an arch lying on its side, and it holds the water by leaning on the rock on both sides of the valley. It is 110 m high. The reservoir holds 379 hm³ and extends into Galicia. One **hm³** (cubic hectometre) is one million cubic metres, that is, one billion litres. It has one **spillway** (descarregador de cheias, the set of gates at the top of the dam used to let large amounts of water through) with 6 **segment gates** (comportas de segmento), which are curved steel doors at the top of the dam that rotate upwards to let water pass. With all 6 gates open, up to 2760 m³/s can pass. It has a **bottom outlet** with a capacity of 400 m³/s. The power plant is inside the mountain and has 2 **generating sets**, 630 MW in total. A generating set (grupo, in dam language) is a turbine together with the generator it drives. The Lindoso sets are of the Francis type, the type used when water falls from a great height. For the interface, the type of turbine does not matter. What matters is that each set can be stopped, starting, generating a certain power (in MW), stopping, or unavailable.

Touvedo dam (Ponte da Barca, 1993). It is a **gravity dam**: it holds the water by its own weight. It is 42.5 m high. The reservoir holds 15.5 hm³, but only 4.5 hm³ are **usable**, that is, only that part can be stored or released by the operator's decision. The rest always stays below the openings. It has 3 segment gates (3200 m³/s with all open), a bottom outlet of 50 m³/s and 1 generating set of the Kaplan type, 22 MW. It receives everything that Lindoso turbines. The two sketches at the end of this document show where the gates, the bottom outlet and the generating sets are in each dam.

Rivers, sensors and forecasts

- **Uncontrolled tributaries:** Vez (enters near Ponte da Barca, coming from Arcos de Valdevez), Vade (coming from Vila Verde) and Estorãos (enters shortly before Ponte de Lima). Each has a **gauging station** (estação hidrométrica), a post on the bank with a sensor that measures the height of the water (the **level**, in metres) and, from it, estimates the flow in m^3/s .
- **Stations on the Lima:** 8 gauging stations (level and flow) and 6 water-quality stations, spread from Lindoso to the river mouth. The quality stations measure **pH** (whether the water is acidic or alkaline), **dissolved oxygen** (the amount of oxygen in the water, which fish need), **conductivity** (which rises when there are salts or pollution) and **turbidity** (how cloudy the water is).
- **Tide and weather:** a **tide gauge** (marégrafo) at Viana do Castelo, a sensor that measures the height of the tide at every moment. The COL also receives the tide table for the following days and the 72-hour rain forecast from IPMA, the Portuguese weather service.

Cameras, treatment-plant discharges and resources

- **Video cameras:** 8, showing live images. They are at the two dams, at the Choupal, at the areal and the Passeio 25 de Abril, at the medieval bridge of Ponte de Lima, at the river mouth and on a riverside road.
- **Treatment-plant discharge points:** a wastewater treatment plant (in Portuguese, ETAR) is where sewage is treated. After treatment, the water is returned to the river at a discharge point. There are three, at Ponte da Barca, Ponte de Lima and Viana do Castelo. They matter to the COL because, if treatment fails, water quality **downstream** (further down the river) drops.
- **Resources:** 4 field teams (2 from the dams, 2 from the Civil Protection), 2 vehicles, warning sirens downstream of the dams, SMS to the parish councils and to the population, and 3 possible closures: i) the Passeio 25 de Abril, ii) the Choupal and Campo da Feira parks, and iii) the municipal roads near the Estorãos.
- **Reference levels:** a reference level (cota) is the height of the water measured from a fixed reference. For each sensitive place there is the level at which the water starts to invade it: the areal, the Passeio 25 de Abril and the Choupal. Each dam also has its safety level, the highest the reservoir may reach. These levels let the interface show, at every moment, how far the water is from each of these limits.

4. States the interface has to show and allow changing

Every element of the system is, at each moment, in a **state**. The interface must always show the current state of each element, in a way the operator understands easily, and must allow changing it when that is permitted. The states below are mandatory. You may add others if you justify them.

- **Segment gate:** closed, opening, open at N metres, closing, blocked, under maintenance, in local manual mode. Local manual mode means that someone at the dam has taken control on the spot, and remote control is no longer available.
- **Bottom outlet:** closed, opening, open, failed.
- **Generating set:** stopped, starting, on load (generating N MW), stopping, unavailable.
- **River stretch:** normal, watch, yellow alert, orange alert, red alert, emergency.
- **Operation** (manobra, any action on gates, bottom outlet or generating sets): requested, awaiting authorisation, authorised, in progress, completed, cancelled.
- **Alert:** new, acknowledged, assigned, being handled, escalated, resolved, false positive.
- **Public warning:** draft, pending, issued, updated, cancelled.
- **Field team:** available, assigned, on the way, on site, returning.
- **Bridge, park or promenade:** open, restricted, closed.

- **Sensor:** ok, suspect, faulty, being calibrated.

5. What you are not expected to do

- Calculate the real hydraulics. You do not need to know how much the river rises if Touvedo releases another 100 m³/s. It is enough that the numbers you invent are plausible against the values in section 3.
- Design the citizen's app. The COL warns the population, but the app the population would use is another project.
- Design EDP's internal system that sends commands to the gate motors and reads the sensors (called SCADA), or the electricity grid. The COL talks to that system, but what you design is what the operator sees and uses.

6. The group's "gold"

All groups work on the same problem. So each group must have **something the others do not have**: an idea, a way of showing, a way of interacting, that solves a hard point well. It does not have to be big. It has to be yours, and it has to be defended with results from the evaluations.

One example, just to give the idea: how to show, in real time, how much room is still left in Touvedo.

7. The room and the equipment

The operator works sitting down, with keyboard and mouse. In front are 2 monitors of 27 inches, UHD, and, further away, a wall of 12 screens (4 by 3) meant to be seen from a distance. The interface must use the three spaces and say what goes on each. Rules:

- It is a critical system. The information the operator needs to decide must be **always visible**, without opening menus and without scrolling.
- There are no hidden menus. Everything that can be done must be in sight or one gesture away.
- A command that changes the state of the river (releasing water, issuing a warning, closing a bridge) must be hard to trigger by mistake and easy to confirm.
- The operator at 3 in the morning, six hours into the shift, must grasp the situation at a glance.

8. How the work develops, week by week

The project is done in the practical classes, in timed tasks, and continued outside them. The Tuesday classes (P1 and P2) and the Friday class (P3) do the same work in the same week. The Fridays of 30 October, 4 December and 11 December are essentially support sessions. There are no classes on 27 October, 1 December and 8 December.

Week	Session	What is done in the practical class	What is finished by the end
22 Sep	S1	Warm-up: short exercises on interfaces and on how to observe the people who use them.	Groups formed.
29 Sep	S2	Guided reading of this statement. Discussion of the problem and of who the operator is. Questions.	Everyone understands the problem and the system.
6 Oct	S3	Each member of the group picks one subject of the system and makes sketches (quick pencil drawings) of how he or she would show and control it on a screen.	One panel per member, in sketch form. Many alternatives, none chosen yet.

13 Oct	S4	With post-its, the group lists everything the operator needs to do and writes the missions (complete tasks, from beginning to end). It ranks the features with the MoSCoW method.	List of missions. List of features with priority.
20 Oct	S5	Consolidation: the group chooses, with written criteria, among the alternatives for each panel and links the panels into a whole. Storyboards of two missions.	A coherent set of sketches and two storyboards.
3 Nov	S6	Phase 1: cognitive walkthrough. Each group goes through one mission, step by step, on another group's sketches and records where a new operator would go wrong.	Walkthrough grid filled in (evidence). List of corrections.
10 Nov	S7	Figma basics. Start of the wireframes: the corrected sketches become screens drawn with precision, on the two monitors and on the wall.	First wireframes in Figma.
17 Nov	S8	Phase 2: heuristic evaluation. Each group examines another group's wireframes against a list of principles of good interfaces and rates the severity of each problem.	List of problems with severity (evidence). Correction plan.
24 Nov	S9	Phase 3: usability tests. Members of another group try to complete missions on the prototype, without help, while time and errors are measured. Homework: repeat with 5 people from outside.	Test protocol and results (evidence).
15 Dec		Final presentations.	

The prototype is evaluated **three times over the semester**, always in class and always by members of other groups. These three phases are part of the grade. What is recorded in each phase is called **evidence**: the filled-in grids, the photographs of the sketches, the evaluators' notes and the test results. The evidence is delivered with the PR2.

Some terms that will appear over these weeks, in order of arrival:

- **Sketch**: a quick pencil drawing, with no concern for precision, to explore ideas.
- **Panel**: an area of the screen, or a whole monitor, devoted to one subject.
- **Mission**: a complete task, from beginning to end, that the operator has to be able to accomplish. Each group writes its own in S4. They must cover normal situations, flood situations and at least one equipment failure.
- **MoSCoW**: a method for ranking features, dividing them into Must (has to exist), Should (ought to exist), Could (might exist) and Won't (left out, for now).
- **Storyboard**: a sequence of frames, like a comic strip, showing the operator accomplishing a mission on the interface.
- **Cognitive walkthrough**: an evaluation in which a mission is followed step by step, asking at each step whether a new operator would know what to do.
- **Wireframe**: a precise drawing of a screen, with the position and size of every element, still without colours, without graphic finish and without behaviour.
- **Heuristic evaluation**: an examination of an interface against a list of principles of good interfaces (the heuristics), with a severity rating for each problem found.
- **Usability test**: watching real people trying to accomplish missions on the prototype, without help, measuring time, errors and what they say.

9. Report 1 (R1)

Due by Sunday 8 November at 23:30, in PDF, 30 MB at most. The R1 tells the group's story up to Phase 1: what you thought, what you drew, how you chose and what the first evaluation showed. It is not a report of pretty design. It is a report of justified decisions. It must contain, in this order:

1. **The problem in your own words** and the three or four biggest interface challenges you identified.
2. **Who operates:** three personas (descriptions of fictional but believable operators, with a name, experience, and what puts them in difficulty). One of them must be an operator with little experience.
3. **The missions** you wrote in S4 (at least five), each with its goal, its starting situation and what must be done by the end. They must cover normal situations, flood situations and at least one equipment failure.
4. **The division of the interface into panels**, with the distribution over the 2 monitors and the wall justified.
5. **The sketches:** all the alternatives you made in S3 (photographed with good contrast), the criteria you used to compare them (for example, how much each helps to accomplish the missions, how much noise it adds, how new the approach is), the scales used, each member's votes and the results. The conclusions of the vote must be written down, and the iterations you made from them shown.
6. **The list of features with priority** (MoSCoW) and the two storyboards.
7. **The cognitive walkthrough (Phase 1):** who evaluated you, the filled-in grid, the problems found and what you decided to change because of them.
8. **The group's "gold"**, as it stands at this moment, even if it is still only an idea starting out.

10. Report 2 (PR2), prototype and evidence

Due by Sunday 13 December at 23:30: the report in PDF (50 MB at most), the link to the Figma prototype (with viewing permission for the lecturers) and the evidence of the three phases in a folder or annex. The PR2 continues the story from the R1 and ends at the final interface. It must contain:

1. **A summary of the R1** in half a page, highlighting what changed in the meantime (if anything did).
2. **The wireframes** and the transition to the Figma prototype. The prototype must be of medium to high fidelity and must allow all the group's missions to be accomplished with real clicks.
3. **The heuristic evaluation (Phase 2):** the list of problems with severity, who found them, and what you changed because of them.
4. **The usability tests (Phase 3):** the protocol (what you asked, of whom, how you measured), the results in class and with the 5 people from outside, and what you changed because of them.
5. **The final interface:** description of each panel, of the states and of how the operator interacts. Clearly readable images.
6. **The group's "gold"**, defended with the results of the evaluations.
7. **Balance:** what was left undone and what you would do differently.

11. Final presentations

On 15 December, in class. Each group shows the prototype working, accomplishing at least three missions live. The presentation must show the path, not just the result: what alternatives you created, how you evaluated them and how the prototype changed because of the evaluations. The time per group depends on the number of groups and is announced in S9.

12. Project assessment

The project grade counts: the R1, the PR2 with the prototype, participation in the three evaluation phases (proven by the evidence) and the final presentation. The weight of each component is published on the course page. A group that misses an evaluation phase without justification loses that component.

13. Glossary

For quick reference. The terms are explained in the text the first time they appear.

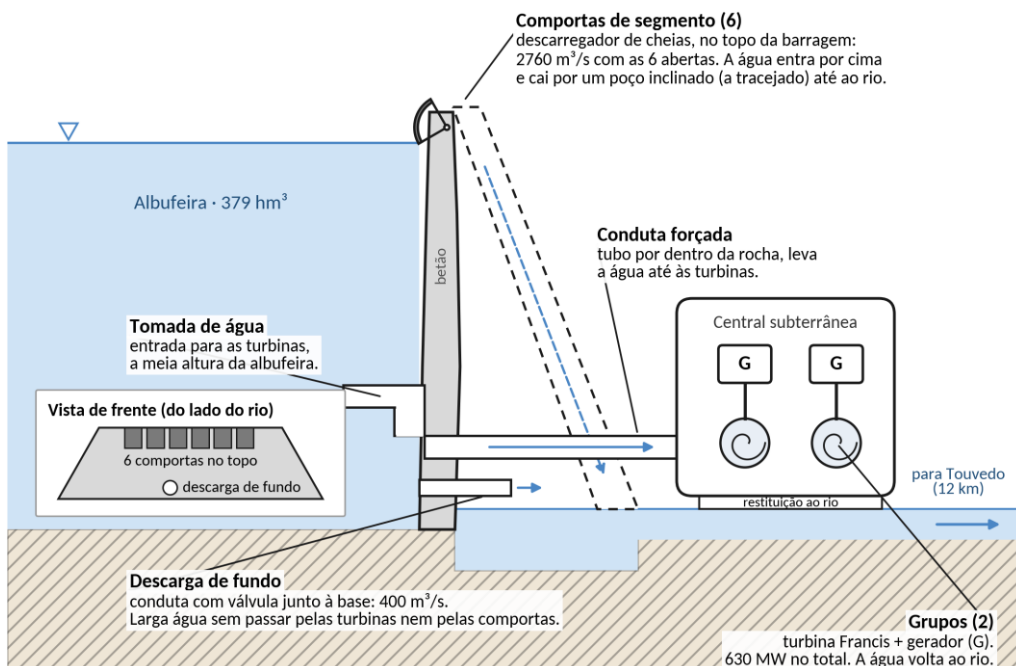
- **Bottom outlet (descarga de fundo):** a pipe with a valve, near the base of the dam, to release water without going through the turbines.
- **Downstream and upstream (jusante e montante):** downstream is down the river, in the direction of the current. Upstream is up the river.
- **ETAR:** wastewater treatment plant. It returns the water to the river, at a discharge point, after treatment.
- **Flow (caudal):** the amount of water passing a point of the river per unit of time, in m^3/s .
- **Gauging station (estação hidrométrica):** a post on the bank that measures the water level and estimates the flow.
- **Generating set (grupo):** turbine plus generator. It produces electricity when water passes through.
- **High tide (preia-mar):** the highest point of the tide. Low tide (baixa-mar) is the lowest.
- **hm^3 :** cubic hectometre, one million cubic metres.
- **Operation (manobra):** any action on gates, bottom outlet or generating sets.
- **Reference level (cota):** the height of the water measured from a fixed reference. The flood level of a place is the level from which the water invades it.
- **Release (largar água):** to let water pass from the reservoir into the river, through gates, bottom outlet or turbines.
- **Reservoir (albufeira):** the artificial lake that forms behind a dam.
- **SCADA:** an industrial system that sends commands to equipment and reads sensors. It is not what you will design.
- **Segment gate (comporta de segmento):** a curved steel door at the top of the dam that rotates to let water through. Also called a radial gate.
- **Shift handover (passagem de turno):** a summary, recorded in the interface, of what happened during the shift and what is left pending, which the next operator reads before taking over the room.
- **Spillway (descarregador de cheias):** the set of gates at the top of the dam, used when a lot of water has to be released.
- **Tide gauge (marégrafo):** a sensor that measures the height of the tide.
- **Tributary (afluente):** a river that flows into another river. Here, the Vez, the Vade and the Estorãos, which flow into the Lima.
- **Turbining (turbinar):** sending water through the turbines to produce electricity.

Annex. Sketches of the two dams

Simplified side views, not to scale, with the three kinds of elements the interface has to show and control: the segment gates, the bottom outlet and the generating sets (turbine plus generator). In each sketch, the small front view shows how many gates there are.

Esboço 1 • Barragem do Alto Lindoso (corte lateral, não à escala)

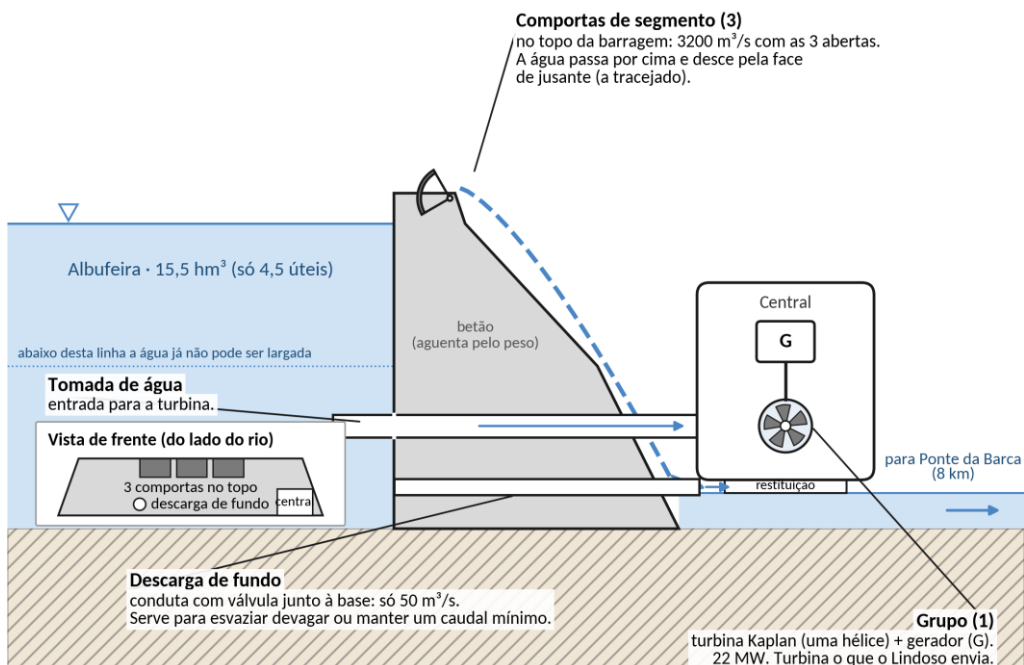
Abóbada com 110 m. À esquerda (montante) a albufeira. À direita (jusante) o rio, que segue para Touvedo.



Sketch 1. Alto Lindoso: 6 segment gates, a 400 m³/s bottom outlet and 2 Francis generating sets in an underground power plant.

Esboço 2 • Barragem de Touvedo (corte lateral, não à escala)

Gravidade com 42,5 m. Recebe o que o Lindoso turbinar e devolve ao rio um caudal regular.



Sketch 2. Touvedo: 3 segment gates, a 50 m³/s bottom outlet and 1 Kaplan generating set in a power plant next to the dam.